

LZ77 Compression and Decompression Report

1. Introduction The purpose of this report is to analyze the implementation of the LZ77 compression and decompression algorithm. LZ77 is a lossless data compression technique that replaces repeating patterns in a string with references to earlier occurrences. This implementation provides an interactive console program that allows users to input text for compression and decompression using LZ77.

2. Algorithm Overview LZ77 operates by scanning the input string and identifying the longest match within a search buffer. The match is then encoded as a tuple (offset, length, next character), where:

- **Position** represents how far back the match starts in the search buffer.
- **Length** is the number of matching characters.
- **Next Character** is the character following the match.

The decompression process reconstructs the original string by using these encoded tuples to retrieve matching sequences from the reconstructed text.

3. Code Implementation The implementation consists of the following components:

- **Tag Class:** Represents the compression tuples containing offset, length, and next character.
- **Compression Method:**
 - Iterates through the input string to find the longest match in a search buffer (size 255 characters).
 - Encodes the match as a tag and moves forward in the string.
- **Decompression Method:**
 - Reads the encoded tags and reconstructs the original string by referencing previous matches.
- **User Interface:**

- Provides options for compression, decompression, and exiting the program.

4. Handling of Repetitive Sequences The program effectively handles repetitive sequences by utilizing a search buffer to detect patterns and encoding them efficiently.

The implementation prevents self-overlapping matches and ensures that decompressed output is identical to the original input.

5. Example Test Cases

Compression Example:

Input: "ababababab"

Output: (0,0,a) (0,0,b) (2,8,null)

Decompression Example:

Input: (0,0,a) (0,0,b) (2,8,)

Output: "ababababab"

6. Conclusion This LZ77 implementation successfully compresses and decompresses text while preserving data integrity. It effectively identifies and encodes repetitive sequences, making it suitable for lossless text compression applications. Future improvements could include optimizing search buffer size and improving match-finding efficiency.